Decentralized Clinical Trial (DCT) design in GLISTEN: a Phase 3 study of linerixibat for cholestatic pruritus in patients with primary biliary cholangitis

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Background

Decentralized Clinical Trials (DCTs)

By definition, a DCT is a clinical trial utilizing technology, processes, and/or services that create the opportunity to reduce or eliminate the need for participants to physically visit a traditional research site.

- DCTs enable choice, more convenience, and access for a wider set of patients to become study participants. They have the potential to enhance inclusion of under-represented populations and they offer innovative and novel study designs to address current challenges of recruitment, engagement, and retention.
- DCTs offer an attractive option for patients who want to participate in clinical research and are of particular use for orphan diseases, where a small number of patients are geographically spread.
- DCTs can be hybrid or fully remote, as they incorporate a range of components to improve the participant's experience.

Organizations including the Decentralized Trials Research Alliance,[1] Clinical Trials Transformation Initiative,[2] TransCelerate BioPharma Inc.,[3] and others have been educating, building awareness to widen acceptance and adoption, and providing evidence to support DCT use.

The US FDA has issued industry guidance for digital health technologies for remote data acquisition in clinical investigations.[4]

Aim

Here we present the study design of GLISTEN and highlight the DCT elements included to improve patient experience.

Possible advantages of DCTs

DCTs in primary biliary cholangitis (PBC): the GLISTEN study

DCTs are of particular relevance to patients with PBC for whom close proximity to a specialist is rare.

Among patients with PBC, 70% of those surveyed indicated they would be more likely to participate in a clinical trial if remote options were available.[5]

Cholestatic pruritus is common in PBC, affecting up to 74% of patients with PBC over their disease course.[6,7]

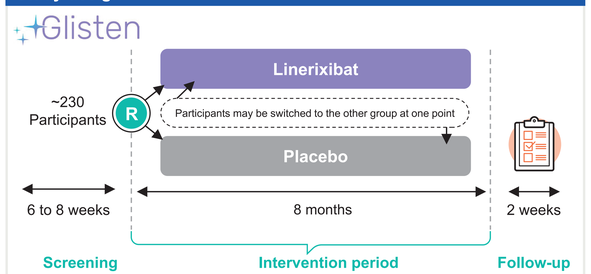
- Current treatments for PBC do not treat cholestatic pruritus. Those used for cholestatic pruritus have limited clinical evidence supporting their use and are often ineffective or poorly tolerated.[8,9]

The Phase 3 GLISTEN trial (NCT04950127) is investigating the efficacy and safety of linerixibat for the treatment of cholestatic pruritus in patients with PBC.

GLISTEN Trial Design

Randomized | Double blind | Phase 3 | Placebo controlled | Multicenter

Study design



Key criteria

Inclusion

- Male and female 18 to 80 years of age
- Participants with proven PBC
- Participants with moderate to severe itch

Exclusion

- Screening total bilirubin >2.0 x ULN
- Screening ALT >6 x ULN
- Use of obeticholic acid within 8 weeks prior to screening
- Initiation, discontinuation or change in dose of UDCA or fibrates in 8 weeks prior to screening

Primary endpoint

Change from baseline

Monthly Itch Scores (the worst week score of the month) over treatment period using 0–10 NRS

Primary

0 No itch — 10 Worst imaginable itch

Every morning: worst itch experienced night before

Every evening: worst itch experienced that day

GLISTEN as a DCT

Several operating models are in place for GLISTEN, including a fully decentralized option

Traditional	Hybrid	Full DCT
Physical sites across all countries	**DCT elements implemented across sites where feasible**	**Single DCT metasite in the USA**
Patients visit site for all study visits	Selected visits and assessments conducted remotely	All study visits are remote, no clinical contact at the study site
Limited patient reach	May promote patient engagement due to reduced burden of site visits	Widest patient reach

Patient reported outcomes data are collected on electronic handheld devices provided by the sponsor across all models

DCT elements included in GLISTEN

Remote screening
Digital recruitment of patients (via social media/patient groups)

eConsent
Electronic form to aid the collection of the informed consent of participant

Telemedicine
Electronic information and telecommunications technologies to support and promote long-distance clinical health care

Electronic clinical outcome assessment
Electronic patient-reported outcome measures

Home health visits
Clinical practitioners provide home nursing visits and phlebotomy remotely in more convenient locations for participants

Direct-to-Participant
Direct shipment of study treatment to the participant's home

Current status

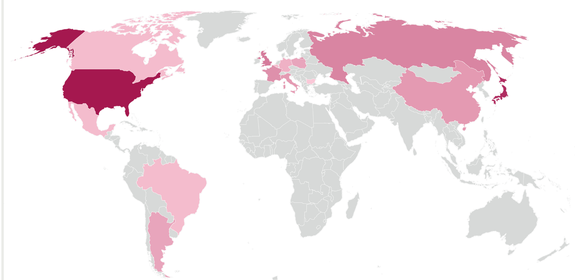
Patient recruitment is ongoing

Primary completion: November 2023

x 84
The study is active at 84 physical study sites worldwide[10]

and as a fully remote DCT in the USA

Phase 3 GLISTEN study locations



Site locations include: United States, Argentina, Belgium, Brazil, Bulgaria, Canada, China, France, Germany, Italy, Japan, Mexico, Poland, Russian Federation, United Kingdom. Number of locations in each country indicated by intensity of red shading.

Conclusions

Incorporation of DCT design elements into GLISTEN is expected to address the issues of limited numbers and diversity of patients recruited, which is particularly important for a rare disease such as PBC.

The GLISTEN study website can be accessed at (https://www.glistentrial.com) or by scanning the QR code.

Disclosures

All authors are employees of and hold shares in GSK.

Abbreviations

ALT, alanine aminotransferase; DCT, decentralized clinical trial; FDA, Food and Drug Administration; NRS, numerical rating scale; PBC, primary biliary cholangitis; R, randomization; UDCA, ursodeoxycholic acid; ULN, upper limit of normal.

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